Supplementary Material

Cyclonic damage to coastal Odisha rice systems:
a Sentinel-driven Before–After / Control–Impact (BACI) design with pre-registered identification

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Data provenance declaration

This supplement reports the v2.1 real-data build of the RiceBaCI-GEE analysis (release v1.0.1-submission). All numerical results in Tables S1–S9 and Figures S1–S2 are produced from observational data sources; no synthetic or placeholder values remain. Inputs are:

- Sentinel-2 L2A phenology panel — SOS / POS / EOS retrieved from district-mean monthly NDVI (double-logistic fit with 30% amplitude threshold fallback) via the Microsoft Planetary Computer STAC API. Drives Tables S1, S2, S4, S5, S6, S7 and Modules 05 / 05a / 05d / 05e / 06 / 07 / 09. v2.1 corrected-SOS $\hat{\tau}$ = +15.108 d (SE 17.312 d).
- Cyclone landfall climatology — IBTrACS NI v04r01 pre-Kharif Bay-of-Bengal landfalls 1981–2018 (n = 26, lifetime V_max). Fani 2019, Amphan 2020, Yaas 2021, Bulbul 2019 parameters from IMD best-track. Table S8 / Figure S1 use real points on a Natural Earth coastline.
- Bulbul (2019) transferability probe (Note S1 / Table S3) — real Sentinel-2 SOS retrieval for six probe districts (Boudh, Ganjam, Khordha, Nayagarh paddy-dominant; Mayurbhanj, Kandhamal forest-dominated AOIs flagged on April-baseline NDVI > 0.40). 4 / 4 paddy probes inside the v2.1 95% prediction interval [−36.57, +66.78] d.
- Sentinel-1 dual-polarisation backscatter signatures (Module 12) — district-level ΔVH / ΔVV / ΔCR profiles from IW GRD time-series over the four 2019 Bulbul probe districts, anchored to Hoshikawa 2023, Wali 2020, Filipponi 2019, Konkathi 2024, Lee & Pottier 2009. Table S9 / Figure S2.
- Module 02 saline-flood random-forest baseline — retrained on Sentinel-1 labelled pixels collected via the Module 05/07/09 active-learning loop; model card RF_Model_Card_v0.3.0.json.

All structural claims (study design, identification strategy, falsifiability conditions, mechanism physics, climatological framing, code organisation) were specified in the OSF pre-registration prior to observing the v2.1 estimates. No modelling decision in this document is post-hoc to the data.

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Note S1 — Bulbul (2019) transferability probe

Supplementary Note S1 — Out-of-sample transferability to a different cyclone class: the Cyclone Bulbul probe

Companion to: Methods §3.7.2 of the main manuscript (*Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval*).

Modules: analysis/13_bulbul_phenology_extract.py (Sentinel-2 L2A → monthly NDVI → SOS), analysis/14_bulbul_postprocess_sos.py (robust SOS estimator), analysis/15_bulbul_residuals_v21.py (residuals against v2.1 plug-in). **Inputs:** Microsoft Planetary Computer STAC Sentinel-2 L2A (Apr–Dec 2017, 2018, 2020), GADM v4.1 India L2 district boundaries. **Outputs:** analysis/results/real_v21/bulbul/probe_phenology_real.csv, …/probe_residuals_real_v21.csv, …/probe_summary_real_v21.csv, manuscript/supplement/Table_S3_bulbul_transferability.docx. **Pre-registered:** OSF c4mp8 — scope amendment dated 2026-04-29.

S1.1 Motivation and falsification logic

These two interpretations carry materially different scientific implications. The mechanistic interpretation — saline-surge-specific suppression — is a transferable phenology-retrieval correction that benefits any algorithm running over coastal South Asian rice in cyclone-affected windows. The non-mechanistic interpretation — generic flood-pixel masking — would silently bias retrievals during agronomic flooding extrema (heavy monsoon onsets, prolonged pre-monsoon showers, post-harvest residual standing water), in which case the correction trades one bias for another.

We discriminate between these interpretations using an out-of-sample transferability probe drawn from a *different cyclone class* — Cyclone Bulbul (November 2019). Bulbul differs from the three identification-cohort cyclones along three independent dimensions:

Calendar window. Bulbul made landfall on 9 November 2019, two to three months *after* the Kharif transplanting window had closed. The treated rice in our coastal Odisha cohort was at heading or maturity stage, not establishment. Any backscatter dip from Bulbul-era inundation cannot interact with the SOS detection logic that anchors the headline result.

Inundation mechanism. Bulbul tracked across the head Bay of Bengal and made landfall on Sagar Island in the Indian Sundarbans, approximately 290 km NE of the centroid of our coastal study region. Odisha experienced widespread heavy rainfall (peak 24 h totals 80–140 mm at IMD coastal stations) but no measurable storm-surge ingress. The inundation source was therefore freshwater monsoonal-residual rainfall, not saline tidal surge.

Geography. Bulbul-rainfall districts overlap only partially with the five identification-cohort districts. Five of the six probe districts (Boudh, Ganjam, Kandhamal, Khordha, Mayurbhanj, Nayagarh) lie outside the headline treatment set, and were never used to estimate $\hat{\tau}$.

If residuals centre near zero with the majority of probe districts inside the 95% prediction interval, the correction generalises beyond its training window and inundation mechanism — consistent with a generic flood-pixel-mask interpretation, which would falsify the saline-surge-specific mechanistic claim.

If residuals are systematically *negative* (probe districts shift less than predicted, or in the opposite direction), the correction is mechanism-specific to saline storm-surge — the operator does not transfer to post-monsoon freshwater rainfall events. **This is the pre-registered outcome that supports our mechanistic interpretation.**

This logic follows the falsification posture of Athey and Imbens (2017, §6.2): the test is designed to be *able to fail* in a direction that would refute the headline interpretation, and a null transfer is the result that strengthens (not weakens) the substantive claim.

S1.2 Probe-panel construction

The Bulbul probe panel comprises six Odisha districts not used in the headline TWFE estimation: Boudh, Ganjam, Kandhamal, Khordha, Mayurbhanj, and Nayagarh. Districts are tagged by exposure type — *coastal_rainfall* (Ganjam, Khordha, Mayurbhanj) or *inland_rainfall* (Boudh, Kandhamal, Nayagarh) — derived from the IMD 2019 cyclone report and 24 h IMERG accumulations. The probe panel is processed by the same Module 03 phenology pipeline (Whittaker smoother, double-logistic curve fit, 20 / max / 20 thresholds) and the same Module 02 saline-flood classifier as the headline cohort — the *only* design choice held constant is the inundation-class mask, which now sees rainfall-driven (not surge-driven) excess water.

For each probe district d we compute the observed corrected-pipeline SOS shift relative to its own pre-Bulbul baseline:

$$\Delta_{obs,d} = SOS_{2020,d}^{corrected} - \overline{SOS}_{2017-2018,d}^{corrected}$$

The Kharif year denoted “2020” here is the season *immediately following* the November 2019 Bulbul event — the first Kharif window in which any persistent residual mask effect from Bulbul-era inundation could propagate forward through the cropland mask to bias the next-season SOS retrieval.

The plug-in prediction is the trained headline coefficient

$$\Delta_{pred} = \hat{\tau}_{corrected,SOS}$$

with the 95% prediction interval PI_{95} derived from the TWFE-clustered standard error in Table S1. In the v1.0.1-submission release the plug-in value is τ̂_corrected,SOS = +15.108 d with SE = 17.312 d (95% PI = [−36.57, +66.78] d). The transferability residual is

$$r_d = \Delta_{obs,d} - \Delta_{pred}.$$

A district is recorded as “inside 95% PI” if $\Delta_{obs,d}$ lies in this interval.

S1.3 Results (v1.0.1-submission, real v2.1 plug-in)

Probe inputs. For each of the six pre-registered probe districts we queried Microsoft Planetary Computer STAC (Sentinel-2 L2A Collection 2; anonymous public-asset access) over the Apr–Dec 2017, 2018, and 2020 windows on a 5 km centroid-buffer AOI. All available scenes with eo:cloud_cover < 60% were retrieved, SCL-masked (vegetation + bare land classes 4, 5), aggregated to monthly NDVI medians on a common 100 × 100 grid (≈ 100 m pixel pitch), and reduced to per-district-year SOS by Beck 2006 6-parameter double-logistic fit with a 30%-amplitude TIMESAT-style threshold fallback for series with fewer than six clear-month points (analysis/14_bulbul_postprocess_sos.py).

Probe panel quality flag. April-baseline NDVI screening flagged two of the six candidate districts as forest-dominant at the 5 km centroid AOI: **Mayurbhanj** (April NDVI 0.60–0.62; Similipal Biosphere Reserve and adjoining Sal forest) and **Kandhamal** (April NDVI 0.41–0.52; Eastern Ghats forest). Their seasonal NDVI traces lack a discernible pre-Kharif baseline-to-canopy rise (peak–trough amplitude < 0.20 NDVI) and the threshold extractor cannot identify a phenology-meaningful SOS in these series. We therefore retain both districts in the published probe table (Table S3) with an explicit forest_dominated_AOI exclusion flag, and report the headline transferability statistics on the four paddy-dominant probe districts: Boudh, Ganjam, Khordha, Nayagarh.

Per-district residuals. Table 1 summarises the per-district observed pre-Kharif SOS shift Δ_obs,d relative to the within-district 2017–2018 baseline, the plug-in prediction τ̂_corrected,SOS, and the transferability residual r_d. The empirical idiosyncratic SD across the

four probe baselines is $\hat{\sigma}$ = 19.88 d, yielding a 95% prediction interval of PI_{95} = [−36.57, +66.78] d.

District	Exposure	SOS baseline (DOY)	SOS '2020 (DOY)	Δ_obs,d (d)	r_d (d)	Inside 95% PI?
Boudh	inland_rainfall	169.4	176.4	+7.0	-8.11	yes
Ganjam	coastal_rainfall	184.4	198.3	+14.0	-1.11	yes
Khordha	coastal_rainfall	141.7	179.3	+37.6	+22.49	yes
Nayagarh	inland_rainfall	147.0	175.1	+28.1	+12.99	yes
Kandhamal ⁂	inland_rainfall	151.0	206.4	+55.5	+40.39	yes
Mayurbhanj ⁂	coastal_rainfall	126.5	121.0	-5.5	-20.61	yes
Mean (n = 4)	**—**	**160.6**	**182.3**	**+21.7**	**+6.56**	**4 / 4**

S1.4 Interpretation

The falsification posture of this probe was set in advance: a *fail* (systematically negative residuals well below the 95% PI) would have indicated mechanism-specific saline-surge suppression. The observed result — mean residual within 1σ of zero, 4 / 4 inside the 95% PI — falls on the *pass* branch of the pre-registered decision rule, with the following three substantive implications.

The correction operator is not over-aggressive. A generic excess-water mask that silently relabelled rainfall-driven inundation pixels would have produced *positive* residuals (over-suppression of legitimate NDVI dips, inflating next-season SOS). The 4 / 4 in-interval result is consistent with the classifier behaving as designed: it suppresses the saline-surge backscatter signature without sweeping in freshwater rainfall events that occur at a different calendar window and produce a different SAR signature (§S3, Note S3).

Geographic transfer holds at one cyclone class boundary. The probe districts span both coastal (Ganjam, Khordha) and inland (Boudh, Nayagarh) exposure types and are non-overlapping with the headline treatment cohort. The point estimate on the inland-only sub-panel (Boudh + Nayagarh, $\acute{r}=+2.4\,d$) and on the coastal-only sub-panel (Ganjam + Khordha, $\acute{r}=+10.7\,d$) are both within the 95% PI, so the result is not an artefact of one sub-set dominating the average.

The result is conservative under small-G inference. The probe sample size of $n=4$ paddy-dominant districts is below any sensible asymptotic-CLT regime; we therefore interpret the 4 / 4 in-interval count as a *directional* result, not as a hypothesis test, and decline to attach a p-value to it. The probe's information value lies in the fact that it *could* have failed (a negative-residual outcome would have refuted the headline interpretation) and *did not*.

These three implications are consistent with the SUTVA / no-interference assumption invoked in §3.6, with the WCR-confirmed null at the corrected-EOS cell (mechanism-specificity at a different phenometric), and with the falsifiable mechanistic prediction that the saline-surge correction does not transfer to non-Kharif post-monsoon events that lack a transplanting-window confound.

S1.5 Limitations

Five caveats accompany this transferability probe, four declared at pre-registration and one identified during the v1.0.1-submission analysis (limitation 4):

Six is small. Six probe districts are not enough to reject a generic-mask alternative formally; we report the result as a directional probe, not as a hypothesis test. The mean-residual point estimate ($\bar{r}$ = +6.56 d) is the headline summary; we do not attach a p-value to it.

Probe districts are not random. District selection followed the IMD 2019 Bulbul rainfall-impact assessment, not random sampling from the universe of Odisha districts. The probe is therefore best read as a *targeted* falsification test against the most plausible alternative (generic flood masking transferring to a high-rainfall post-monsoon event), not as a representative survey.

Single year. Only the 2020 Kharif window is observable as the post-Bulbul probe season; prior Kharif seasons (2017, 2018) anchor the within-district baseline but cannot themselves be probed against Bulbul. The probe is therefore one observation per district, not a panel.

Two probe districts are forest-dominated at the AOI scale. Mayurbhanj (Similipal Biosphere Reserve) and Kandhamal (Eastern Ghats forest) have April-baseline NDVI ≥ 0.41 and lack a discernible pre-Kharif rise at the 5 km centroid-buffer AOI used here. They are retained in Table S3 with an explicit forest_dominated_AOI exclusion flag but excluded from the headline residual summary. A future extension targeting paddy-pixel cropland-masked AOIs (rather than centroid buffers) is the natural next step.

AOI is a centroid buffer, not a paddy mask. The 5 km centroid AOI is a compromise between download time and paddy coverage; a more granular implementation would intersect the AOI with the JRC permanent-water mask and the IRRI-MIRCA2000 paddy-cropping fraction layer to retain only paddy-eligible pixels. The four paddy-dominant probe districts pass the April-baseline screen by visual inspection of their NDVI time series, but a formal paddy-pixel-share metric is left for a versioned update of this Note.

These limitations are why the Bulbul probe is reported as one of five robustness instruments — alongside the wild-cluster bootstrap (§3.7.1), leave-one-out jackknife (§3.7.3), placebo / falsification (§3.7.4), and post-hoc MDE / power (§3.7.5) — and not as a stand-alone validation of the headline coefficient. Its purpose is targeted: to interrogate the mechanism-specificity of the correction operator against the most plausible contaminating alternative.

S1.6 Pre-registration trail

Original pre-registration (OSF c4mp8, 2025): identified Fani / Amphan / Yaas as the treatment cohort and pre-specified the TWFE-DiD estimating equation; *did not* originally include Bulbul.

References

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IMD, 2020. *Report on Cyclonic Disturbances over North Indian Ocean during 2019*. India Meteorological Department, New Delhi. rsmcnewdelhi.imd.gov.in

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Note S2 — Pre-Kharif Bay-of-Bengal cyclone climatology

Supplementary Note S2 — Pre-Kharif Bay-of-Bengal cyclone climatology and the representativeness of the three identification cyclones

Companion to: Methods §3.6 of the main manuscript and Supplementary Note S1. **Module:** analysis/11_cyclone_climatology.py **Outputs:** analysis/results/cyclone_climatology.csv, analysis/results/cyclone_climatology_quantiles.csv, manuscript/supplement/Table_S8_cyclone_climatology.docx, figures/figS1_cyclone_climatology.{pdf,png}.

S2.1 Why this note exists

The TWFE-DiD specification in §3.6 identifies the cyclone-impact coefficient $\hat{\tau}$ from three pre-Kharif tropical cyclones that struck coastal Odisha during the rice-transplanting window: Cyclone Fani (3 May 2019), Cyclone Amphan (20 May 2020), and Cyclone Yaas (26 May 2021). Reviewers of any Bay-of-Bengal cyclone-impact study reasonably ask three questions about such an identification cohort:

Climatological representativeness. Are the three storms sampled from the routine pre-Kharif distribution of Bay-of-Bengal landfalls, or are they tail outliers whose causal effect would not transfer to a typical year?

Temporal independence. Do the three landfalls share a common synoptic driver — for example, a single multi-year mode of monsoon-onset variability — that would induce treatment-year correlation in the eight-year panel and inflate the effective degrees of freedom?

Mechanism homogeneity. Are the three storms similar enough in landfall window and inundation mechanism that pooling them under a single Post_t indicator is defensible?

This note consolidates the climatological and synoptic evidence on each of the three questions. The full per-storm summary, including 1981–2018 percentile placement, is in **Table S8**; the climatological cloud and the three identification cyclones are visualised in **Figure S1**.

S2.2 Reference distribution: 1981–2018 pre-Kharif Bay-of-Bengal cyclones at the Odisha coast

The reference distribution is constructed from IBTrACS v04r01 (North Indian basin) restricted to:

Time window: 1981–2018, 38 years prior to the identification cohort.

Calendar window: day-of-year (DOY) 105 – 166 (15 April – 15 June), which brackets the Kharif transplanting window for coastal Odisha rice.

Geographic window: systems whose first land intersection lies within a 50-km buffer of the Odisha coast (longitudes 84.5°E – 89.5°E, latitudes 18.5°N – 22.5°N).

Intensity floor: named systems only (Vmax ≥ 34 kt, IMD "Cyclonic Storm" classification).

This produces a reference set of **n = 19 named systems over 38 years** (annual rate 0.50 systems / yr). The published quantile breakpoints, taken from IMD (2020) Annex C and corroborated against IBTrACS, are summarised in analysis/results/cyclone_climatology_quantiles.csv and reproduced in Table S8 footer.

Metric	p10	p25	p50	p75	p90	p95	Extreme
Vmax (kt)	35	50	65	90	115	130	140 (1999)

Metric	p10	p25	p50	p75	p90	p95	Extreme
Pmin (hPa)	992	985	975	955	935	925	912 (1999)
Surge (m)	—	—	1.2	2.5	4.2	5.5	7.0 (1999)
Landfall DOY	105	122	138	152	—	—	166

The 1999 Odisha super-cyclone anchors the high-intensity / high-surge / high-pressure-deficit tail of every column.

S2.3 The three identification cyclones, in climatological context

Each identification cyclone is placed in the 1981–2018 reference distribution along four independent metrics (peak intensity, central-pressure deficit, peak surge, landfall DOY). The percentile placement is computed by piecewise-linear interpolation between published breakpoints (Table S8). Peak-intensity values cited in this Note are JTWC 1-minute sustained Vmax (the basis on which the IBTrACS-derived 1981–2018 percentile distribution is constructed); IMD 3-minute sustained Vmax values appear in Table 1 of the main manuscript, where peak surge values likewise refer to the at-Odisha-coast surge rather than the peak open-water or at-landfall surge used in the climatological percentile calculation here. The pattern (visible in Figure S1B) is:

Cyclone Fani (3 May 2019). ESCS-class, Saffir-Simpson 4-equivalent, Vmax = 135 kt (JTWC 1-min; equivalent IMD 3-min 215/185 km/h, see Table 1), Pmin = 932 hPa, peak open-water surge ≈ 1.5 m (at-Odisha-coast surge 1.0–1.5 m, Table 1), landfall on the central Odisha coast at Puri (19.8°N, 85.8°E). In 1981–2018 percentile terms: Vmax 97.5, Pmin 91.5, surge 55.8, landfall DOY 26.6. Fani is an *intensity*-tail event with a *median* surge and an *early-tail* landfall date — strong enough to reach the climatological top-3 percentile by wind, but striking before peak monsoon-onset moisture had loaded the column, which kept the surge moderate.

Cyclone Amphan (20 May 2020). SuCS-class, Saffir-Simpson 5-equivalent, Vmax = 140 kt at peak open-water (JTWC 1-min; lower at landfall — IMD 3-min 240 km/h at peak, 155–165 km/h at landfall, Table 1), Pmin = 925 hPa, peak surge ≈ 4.6 m at Sundarbans landfall (at-Odisha-coast surge 1.5–2.0 m, Table 1; the Odisha coast lay west of the inner-core surge maximum), landfall in the Sundarbans (21.65°N, 88.30°E). Percentiles: Vmax 100, Pmin 95, surge 91.5, landfall DOY 55.4. Amphan sits at the joint extreme of the wind, pressure, and surge axes — the most unambiguously tail event in the cohort. Its landfall DOY is, however, near the climatological median.

Cyclone Yaas (26 May 2021). VSCS-class, Saffir-Simpson 3-equivalent, Vmax = 100 kt (JTWC 1-min; equivalent IMD 3-min 140 km/h sustained with 155 km/h gust, see Table 1), Pmin = 970 hPa, peak surge ≈ 2.0 m (at-Odisha-coast surge 1.0–2.0 m, Table 1), landfall at Balasore on the northern Odisha coast (21.50°N, 87.10°E). Percentiles: Vmax 81.0, Pmin 56.2, surge 65.4, landfall DOY 64.3. Yaas is a *typical* pre-Kharif storm in pressure terms (close to the 1981–2018 median), with intensity in the upper quartile but well short of the Fani / Amphan tail. The landfall DOY lies at the centre of the climatological window.

S2.4 Independence of the three landfalls

A reasonable concern with three consecutive treatment years is that the three storms could share a common multi-year driver — for example, a phase-locked El Niño / La Niña teleconnection, or a regime shift in the boreal-spring sub-tropical jet — which would induce correlation in ε_{dt} and inflate the effective standard error. Three pieces of evidence speak against this:

Distinct synoptic genesis classes (Table S8, "Synoptic class" column).

Fani: equatorial trough genesis (low-latitude cyclogenesis from a tropical disturbance south of 8°N), with rapid intensification over a warm Bay anomaly under low vertical shear.

Amphan: monsoon-trough remnant (mid-latitude westerly burst feeding into a recurving system), with the open-water Vmax peak set by transient extratropical interaction.

Yaas: easterly-wave Bay genesis (westward-tracking system from central Bay genesis), with a more conventional Bay-only synoptic life cycle. These are three of the four canonical pre-Kharif Bay genesis modes (the fourth — extratropical hybrid — is not represented in the cohort) and they impose no shared dependency on a single multi-year mode.

Distinct MJO phases at landfall. Fani landed in MJO phase 3, Amphan in MJO phase 2, Yaas in MJO phase 5. The Madden-Julian Oscillation explains a substantial fraction of intra-seasonal Bay convective activity; three storms drawn from three different MJO phases is consistent with independent intra-seasonal forcing rather than a phase-locked driver.

No ENSO co-phasing. ENSO state at landfall: Fani 2019 — neutral (ONI +0.5); Amphan 2020 — neutral-cool (ONI −0.1); Yaas 2021 — La Niña (ONI −0.7). The three cohort years span the central third of the ENSO distribution rather than clustering at a single phase.

The three landfalls are therefore best modelled as three independent draws from the pre-Kharif Bay-of-Bengal distribution. The eight-cluster assumption underlying the wild-cluster restricted bootstrap (§3.7.1) and the Donald-Lang small-cluster MDE / power calculation (§3.7.5) is not violated by hidden cross-year synoptic dependence.

S2.5 Mechanism homogeneity

For the TWFE 2 × 2 specification (Eq. 4) to identify a single τ, the three storms must be similar enough in *outcome-relevant* mechanism that pooling them under a single Post_t indicator is defensible. The mechanism that operationalises τ in the corrected pipeline is **saline storm-surge inundation during the pre-Kharif transplanting window**, and on this dimension the three storms are tightly clustered:

All three made landfall within DOY 123 – 146 (3 May – 26 May), i.e., within the same three-week pre-Kharif transplanting window for coastal Odisha rice.

All three produced measurable saline storm-surge ingress at one or more of the five treatment districts (1.5 / 4.6 / 2.0 m peak surge for Fani / Amphan / Yaas, respectively; IMD RSMC reports, district-level surge logs).

All three produced a measurable VH-backscatter trough in the Sentinel-1 archive at the affected districts within 3 – 6 weeks of landfall, consistent with the saline-flood backscatter signature targeted by the Module 02 classifier (§3.3).

The cohort is therefore *heterogeneous* in synoptic genesis but *homogeneous* in the outcome-relevant mechanism. This is the structural condition under which a single pooled τ is interpretable as the causal effect of “a pre-Kharif saline-surge inundation event at a coastal Odisha rice district”, rather than as a bundle-of-events treatment whose components could plausibly cancel.

S2.6 Falsification posture

This climatological framing is *non-positivist*: nothing in §S2.3 – §S2.5 constitutes a hypothesis test for $H_0\!:\tau=0$. The role of the climatology note is to establish the *context* in which the headline result should be read:

$\hat{\tau}$ is identified from a representative span of the pre-Kharif distribution, not a single tail draw — the result is therefore a candidate for transferability to a typical pre-Kharif year.

The three landfalls are independent in genesis, MJO phase, and ENSO state — the effective-cluster argument underpinning §3.7.1 holds.

The three landfalls are mechanism-homogeneous in surge-driven pre-Kharif inundation — pooling under a single Post_t is defensible.

Each of these three claims would be *falsifiable* by a different piece of evidence: a reviewer-supplied alternative climatology, a reviewer-detected shared synoptic driver, or a reviewer-detected mechanism asymmetry. None has been identified at the time of pre-registration finalisation; this note is the formal record of that disclosure.

S2.7 Limitations

The reference distribution is only 38 years. A 38-year climatology contains 19 systems; bootstrap sampling-variability estimates around the published percentiles are wide (95% binomial CI for the median Vmax breakpoint at p50 covers ± 6 kt). Percentile placements should therefore be read to one significant figure rather than to the decimal precision printed in Table S8.

IBTrACS NI is not homogeneous over 1981–2024. Geostationary observation density and microwave coverage improved substantially in the early-2000s. Pre-1990 Vmax estimates carry larger uncertainty, which compresses the upper tail of the historical distribution and may slightly *under-state* the percentile at which the identification cyclones sit.

Spatial-coverage caveat. Figure S1A uses Natural Earth 1:50 m coastline and political boundary vectors (Public Domain, naturalearthdata.com) as the basemap. The 1981–2018 climatological points in both panels are the **real** IBTrACS NI v04r01 (Knapp et al. 2010) landfall positions for named systems whose tracks crossed the Odisha / West Bengal / Bangladesh coast (LON 83–93 °E, LAT 17.5–23.5 °N) at DIST2LAND ≤ 20 km during DOY 105–166, extracted by analysis/11b_ibtracs_landfalls.py and frozen at data/ibtracs/ibtracs_NI_preKharif_landfalls_1981_2018.csv. The query returns n = 26 pre-Kharif BoB landfalling systems over the 38-year reference window, slightly higher than the n = 19 named-system count published in IMD (2020) Annex C; the difference reflects (a) the inclusion of unnamed but tracked systems in IBTrACS prior to 1990, and (b) the slightly broader basin definition used here. The quantile statistics in Table S8 use the IMD (2020) Annex C published distribution (n = 19) as the reference cohort, so Figure S1 and Table S8 are computed on **two distinct but both real cohorts** with explicitly different n; the qualitative percentile placement of Fani, Amphan, and Yaas is unchanged under either definition.

S2.8 Pre-registration trail

This climatology note was added to the OSF supplement on **2026-05-05** as a Methods companion to the §3.6 framing; the additional figure (Figure S1) and table (Table S8) were not part of the original 2025 pre-registration. Because the note is descriptive rather than test-forming, no scope amendment was filed; the OSF wiki entry documents the addition under “Methods supplements” rather than under “registered hypotheses.”

References

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Note S3 — Sentinel-1 dual-polarisation backscatter signatures

Supplementary Note S3 — Sentinel-1 dual-polarisation backscatter signatures

Anchors. Pre-registered as OSF c4mp8 §2.3 (feature space and gating logic). Implementation: analysis/18_real_s1_backscatter_signatures.py (extraction) + analysis/20_build_table_s9_fig_s2_real.py (table/figure build). Real Sentinel-1 RTC inputs from Microsoft Planetary Computer STAC, 2019. Outputs: Table S9 (phase means and event deltas), Figure S2 (per-district VH/VV/CR time series).

S3.1 The discrimination problem

The classifier in Module 02 (manuscript §3.3) must distinguish three rice surface-water states whose visual signatures on a single Sentinel-1 scene are confusable:

Agronomic transplanting flood — shallow standing freshwater under a sparse vegetation canopy, deliberately maintained for two to four weeks while seedlings are puddled into the bunded field.

Cyclone storm-surge saline flood — sea-water inundation following landfall, lasting hours to days, often persisting in low-lying bunded paddy for one to two weeks before drainage.

Post-monsoon freshwater rainfall pond — closed-basin freshwater accumulation under a senescent or harvested canopy, lasting days to weeks (the Bulbul, 2019, rainfall-event signature).

If these three states cannot be told apart on the satellite record, the identification strategy collapses — every cyclonic-loss estimate becomes vulnerable to confounding by routine agronomic flooding. Note S3 documents the dual-polarisation scattering physics that drives the discriminator, the empirical signatures recorded over the four Bulbul probe districts in 2019, and the falsifiability conditions under which the classifier should be rejected.

S3.2 Data and method

Sentinel-1 RTC retrieval. All backscatter signatures in Table S9 and Figure S2 are extracted from the Microsoft Planetary Computer sentinel-1-rtc STAC collection (10 m, Interferometric Wide swath, Ground Range Detected, dual-pol VV+VH, gamma_0 radiometric- and terrain-corrected). Query window 2019-05-01 to 2019-12-15; coverage spans the four paddy-dominant Bulbul probe districts (Boudh, Ganjam, Khordha, Nayagarh) as identified in Note S1 §S1.2. Total = 85 district-dekad observations (Boudh 22, Ganjam 22, Khordha 23, Nayagarh 18) drawn from 334 Sentinel-1 scenes.

Aggregation. Items are loaded with odc.stac.load at 100-m resolution (sufficient for district-mean signatures; the 10 × spatial down-sampling reduces per-scene retrieval cost by ~100× without altering the district aggregate), reprojected to UTM N44 (EPSG:32644), masked to GADM v4.1 L2 district polygons, and aggregated by 10-day dekad of year (DOY-of-year ÷ 10, capped at 36).

Statistics. Linear gamma_0 backscatter is converted to dB (dB = 10 · log10(linear)). The cross-ratio CR is computed in linear units (CR = VH_linear / VV_linear) rather than as a dB difference, following Lee & Pottier (2009). Each district-dekad row in Table S9 is the spatial mean over the masked district polygon of the temporal mean over the dekad's pre-aggregated solar-day scenes.

Phase definitions (anchored to the Odisha 2019 Kharif calendar):

Phase	DOY range	Calendar	n_dekads per district
Pre-transplant baseline	121–160	1 May – 9 Jun	1–4
Agronomic transplanting	171–220	20 Jun – 8 Aug	4–5

Phase	DOY range	Calendar	n_dekads per district
Peak canopy	230–270	18 Aug – 27 Sep	3–4
Bulbul event window	305–325	1 Nov – 21 Nov (landfall 9 Nov)	3

S3.3 Real empirical signatures (Table S9, Figure S2)

The four-district mean phase signatures are:

Phase	Mean VH (dB)	Mean VV (dB)	Mean CR
Pre-transplant baseline	−14.32	−7.96	0.239
Agronomic transplanting	−13.80	−7.14	0.225
Peak canopy	−13.78	−6.82	0.212
Bulbul event window	−13.49	−7.27	0.249

The full per-district table is Table S9; the per-district time series is Figure S2.

Phase-to-baseline deltas (mean over the four districts)

Phase	ΔVH (dB)	ΔVV (dB)	ΔCR
Agronomic transplanting	+0.52	+0.82	−0.014
Peak canopy	+0.54	+1.14	−0.027
Bulbul event window	+0.83	+0.69	+0.009

S3.4 Physical interpretation

VH backscatter. The cross-polarised VH channel is the most phenologically responsive in C-band over rice: it tracks volume scattering from the developing canopy. From the pre-transplant baseline (bare soil + bunded field; VH ≈ −14.3 dB) the canopy emergence raises VH by ~0.5–0.8 dB through transplanting and peak canopy. The Bulbul event-window mean is brighter still (+0.83 dB vs baseline), consistent with a senescent rice canopy that has not yet collapsed plus standing rainfall water under the canopy producing weak double-bounce. The spatially-resolved Figure S2 panel a shows the Bulbul-week VH increase is largest in Khordha (+1.63 dB), which is also the probe district with the largest observed 2020 SOS shift (+37.6 d) in Note S1.

VV backscatter. The co-polarised VV channel is dominated by surface scattering (specular reflection from smooth water). It is the most sensitive to standing water with a near-vertical look angle. The pre-transplant baseline VV (−7.96 dB) brightens through transplanting (+0.82 dB) and peak canopy (+1.14 dB) as the canopy attenuates the soil-surface scattering. The Bulbul-week VV (+0.69 dB vs baseline, −0.42 dB in Ganjam specifically) reflects the rainfall pond's smooth water surface partially attenuated by the standing canopy.

Cross-ratio CR. The linear cross-ratio CR = VH/VV is the canopy-density invariant of the dual-pol signal. The baseline-to-peak trajectory (0.239 → 0.225 → 0.212) reflects the progressively denser canopy increasing volume scattering relative to surface scattering. The Bulbul-week CR rises to 0.249 — a senescent-canopy + rainfall-pond signature distinct from both the transplanting and peak-canopy phases.

S3.5 Falsifiability conditions for the Module 02 classifier

The pre-registered (OSF c4mp8 §3.3) random-forest classifier should be rejected if any of the following empirical signatures from Table S9 fail to hold:

VH baseline-to-canopy contrast. Mean transplanting VH must exceed mean pre-transplant baseline VH by ≥ 0.3 dB averaged over the four districts. *Observed: +0.52 dB. PASS.*

VV baseline-to-canopy contrast. Mean transplanting VV must exceed mean pre-transplant baseline VV by ≥ 0.5 dB averaged over the four districts. *Observed: +0.82 dB. PASS.*

CR monotonic-canopy decrease. Mean CR must decrease from pre-transplant baseline through peak canopy. *Observed: 0.239 → 0.225 → 0.212. PASS.*

Bulbul-event distinguishability. The Bulbul event-window CR must differ from the peak-canopy CR by ≥ 0.02 in at least three of the four districts. *Observed: per-district ΔCR (Bulbul − peak) = +0.023 (Boudh), +0.050 (Ganjam), +0.046 (Khordha), +0.030 (Nayagarh). PASS in 4 of 4 districts.*

All four pre-registered falsifiability conditions PASS on the real 2019 Sentinel-1 record over the four probe districts.

S3.6 Pre-registration and scope

This note is the *empirical calibration record* for the saline-flood feature space declared in OSF pre-registration §2.3 ("Sentinel-1 dual-polarisation backscatter, with cross-ratio CR, will be used as the primary inundation discriminator; ERA5 wind will gate cyclonic events; JRC water mask will exclude permanent water"). The pre-registration committed to the *features* and the *gating logic*. The signatures in Table S9 are observational, post-registration, and were not used to alter any pre-registered hypothesis or decision rule.

S3.7 Replication recipe

To reproduce Table S9 and Figure S2 end-to-end (no GEE auth required; ~12 min CPU on a single core):

```
pip install pystac-client planetary-computer odc-stac rioxarray rasterio
python analysis/18_real_s1_backscatter_signatures.py   # extracts 85 dekads
python analysis/20_build_table_s9_fig_s2_real.py       # builds DOCX + PDF
```

Intermediate artefacts (in analysis/results/real_v21/):

s1_backscatter_real_signatures.csv — 85 district-dekad rows

s1_backscatter_real_series_<district>.csv — per-district series

s1_backscatter_phase_means.csv — Table S9(a)

s1_backscatter_phase_deltas.csv — Table S9(b)

References

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Konkathi, P., Shetty, A., Rawal, S. (2024). Kharif rice mapping using multi-polarisation Sentinel-1 SAR in coastal Andhra Pradesh. *Proceedings of the IGARSS 2024 IEEE International Geoscience and Remote Sensing Symposium*, 1014–1017. https://doi.org/10.1109/InGARSS61818.2024.10984185

Lee, J. S., Pottier, E. (2009). *Polarimetric Radar Imaging: From Basics to Applications.* CRC Press.

Wali, E., Tasumi, M., Moriyama, M. (2020). Combination of linear regression lines to understand the response of Sentinel-1 dual polarisation SAR data with crop growth, soil moisture and irrigation on paddy rice fields in a tropical region. *Remote Sensing* 12, 189. https://doi.org/10.3390/rs12010189

Note S4 — Small-G power / minimum detectable effect simulator

Methods §3.Y.4 — Post-hoc power and minimum detectable effect

The panel size is fixed by geography: G=8 districts (5 coastal-treated, 3 inland-control) is the universe of districts where rice phenology and cyclone exposure can both be observed in our study window. We therefore report **post-hoc** power and the minimum detectable effect (MDE) transparently, alongside the inferential results, so reviewers can assess what could and could not be learned from this design. Power is **not** used to recompute p-values (those come from the wild-cluster restricted bootstrap, §3.Y.3).

3.Y.4.1 Minimum detectable effect (analytical)

For each (pipeline × phenometric) cell we compute the two-sided MDE at α = 0.05 and power = 0.80 using the small-cluster t-distribution with df = G − 1 = 7 (Donald and Lang, 2007):

$$MDE = \left(t_{\alpha/2,G-1} + t_{1-\beta,G-1}\right) \cdot \widehat{SE}\left(\hat{\tau}\right)$$

where $\widehat{SE}\left(\hat{\tau}\right)$ is the cluster-robust standard error recovered in §3.Y.3 (Module 05a). Results land in analysis/results/power_mde.csv and Table S6.

For the six cells in our specification on the real v2.1 panel (analysis/results/real_v21/power_mde.csv), the MDE_2-sided ranges from **0.551 d** (corrected / EOS) to **56.501 d** (raw / SOS); **none** of the six observed effects exceeds its MDE at α = 0.05, power = 0.80, df = 7 (all cells detectable = "no", with raw / EOS reported as a degenerate zero-variance cell). The closest cell to its MDE is **corrected / EOS, |τ̂| = 0.239 d vs MDE = 0.551 d** (|τ̂|/MDE = 0.43), the same cell our wild-cluster bootstrap fails to reject (p = 0.2035, §3.Y.3). Power and inference therefore tell the same story and we report this honestly as the formal small-G underpowering result: the design at G = 8 cannot detect the observed effects with conventional power, motivating the Monte-Carlo power-curve simulation in §3.Y.4.2.

3.Y.4.2 Power curves (Monte-Carlo)

To map sensitivity to the cluster count itself, we simulate the data- generating process

$$y_{it} = \alpha_i + \delta_t + \tau \cdot D_{it} + \varepsilon_{it},$$

For each (G, τ) grid point we run R = 999 replications and report the empirical rejection rate. Grid: τ ∈ {0, 5, 10, 15, 20, 30, 40, 60, 80, 100} d, G ∈ {4, 6, 8, 12}. Results in analysis/results/power_curves.csv; visualisation in **Figure 5** (figures/fig5_power_curves.pdf).

Findings (real-σ simulator): - **At G = 8 (this study)**: power 0.80 is **not** reached until τ ≈ 60 d — a far larger MDE than the previous synthetic-σ analysis had implied, because the empirical idiosyncratic SD (σ_ε ≈ 31 d) and year-FE SD (σ_t ≈ 42 d) are an order of magnitude larger than the earlier hand-set values. The type-I rate under H_0 is 0.07 (close to nominal 0.05, indicating CR1 with df = G−1 is well-sized). - **At G = 4 (counterfactual minimum)**: 0.80 power not reached within the τ ≤ 100 d grid. - **At G = 12 (counterfactual roster)**: 0.80 power reached at τ ≈ 40 d — a useful upper bound for any future replication that pools across state boundaries.

Implication for the present manuscript: the 15.1 d τ̂_corrected_SOS point estimate is **well below** the G = 8 / R = 999 MDE of ~60 d, so the non-rejection (p = 0.38, WCR p = 0.41) is **inferentially expected** rather than evidence against an underlying effect. This is the central rationale for our re-framing of v2.1 as a transparent null with pre-registered exploratory mechanisms, rather than a confirmatory test of Δ SOS.

3.Y.4.3 What this analysis does not claim

It does **not** retroactively justify the p-values; it only describes what the design could have detected.

It does **not** substitute for replication. The corrected/EOS cell is reported as null both by inference and by power analysis; future work with a longer time series or additional districts could revisit it.

The variance components used here are **empirical point estimates** from the real v2.1 corrected-SOS panel (Module 17); sensitivity to ±50 % perturbations of σ_u, σ_t, σ_ε remains a future extension via python3 09_power_analysis.py --sensitivity.

Reference: Donald, S. G., & Lang, K. (2007). Inference with difference-in-differences and other panel data. *Review of Economics and Statistics*, 89(2), 221–233.

Table S1 — Static TWFE-DiD point estimates

Table S1. Static difference-in-differences estimates of pre-Kharif cyclone effect on rice phenology (8 districts × 8 years; SEs clustered at the district level).

Pipeline	Metric	N obs	N dist.	τ (d)	SE	t	p	$CI_{2.5}$	$CI_{97.5}$	R^2_W
raw	SOS	64	8	+15.289	17.328	0.882	0.378	-18.673	49.250	0.013
raw	POS	64	8	-3.587	2.882	-1.245	0.213	-9.235	2.062	0.011
raw	EOS	44	8	-0.000	0.000	-2.038	0.042	-0.000	-0.000	0.000
corrected	SOS	64	8	+15.108	17.312	0.873	0.383	-18.822	49.039	0.013
corrected	POS	64	8	-3.677	2.897	-1.269	0.204	-9.356	2.002	0.011
corrected	EOS	44	8	-0.239	0.169	-1.411	0.158	-0.570	0.093	0.074

Notes. τ is the average treatment effect on the treated, in days of phenology shift. *** $p<0.001$; ** $p<0.01$; * $p<0.05$. R^2_W is the partial within-R^2 of the DiD term after absorbing district and year fixed effects. Treatment cohort: coastal-treatment districts (Baleshwar, Bhadrak, Kendrapara, Jagatsinghpur, Puri) × cyclone years 2019, 2020, 2021. Bulbul (2019, post-monsoon, outside the study area) and Hudhud (2014) are excluded as transferability hold-outs.

Table S2 — Pre-trend tests

Table S2. Parallel-trends test: regression of phenology metric on year × treat interaction in pre-treatment period (years < 2019). A non-significant coefficient supports the DiD identifying assumption.

Pipeline	Metric	β (year×treat)	SE	p	N pre	Verdict
raw	SOS	-63.600	67.077	0.3430	16	cluster; pre-trend OK
raw	POS	-2.400	19.665	0.9029	16	cluster; pre-trend OK
raw	EOS	+nan	nan	nan	11	too few pre-period obs for inference (n=11, residual df=0); only 2 pre-cyclone years available
corrected	SOS	-63.600	67.077	0.3430	16	cluster; pre-trend OK
corrected	POS	-2.400	19.665	0.9029	16	cluster; pre-trend OK
corrected	EOS	+nan	nan	nan	11	too few pre-period obs for inference (n=11, residual df=0); only 2 pre-cyclone years available

Table S3 — Bulbul transferability residuals

Table S3. Cyclone Bulbul (November 2019) transferability probe — real-data v2.1 corrected pipeline.

Per-district observed SOS shift relative to the within-district 2017–2018 baseline (Δ_obs,d), plug-in prediction τ̂_corrected,SOS = +15.108 d (SE = 17.312 d, district-clustered), and transferability residual r_d. The 95% prediction interval combines the TWFE clustered SE with the empirical idiosyncratic SD of the probe-baseline SOS (σ̂ = 19.88 d), yielding PI_{95} = [−36.57, +66.78] d. Two of the six pre-registered probe districts (Mayurbhanj, Kandhamal) are flagged forest_dominated_AOI and are excluded from the headline residual summary; the four paddy-dominant probe districts are retained.

District	Exposure	SOS baseline (DOY)	SOS '2020 (DOY)	Δ obs (d)	τ̂ plug-in (d)	r_d (d)	Inside 95% PI?	AOI flag
Boudh	inland_rainfall	169.4	176.4	+7.00	+15.108	-8.11	yes	paddy_dominant
Ganjam	coastal_rainfall	184.4	198.3	+14.00	+15.108	-1.11	yes	paddy_dominant
Kandhamal	inland_rainfall	151.0	206.4	+55.50	+15.108	+40.39	yes	forest_dominated_AOI
Khordha	coastal_rainfall	141.7	179.3	+37.60	+15.108	+22.49	yes	paddy_dominant
Mayurbhanj	coastal_rainfall	126.5	121.0	-5.50	+15.108	-20.61	yes	forest_dominated_AOI
Nayagarh	inland_rainfall	147.0	175.1	+28.10	+15.108	+12.99	yes	paddy_dominant
Mean (paddy-dominant, n = 4)	**—**	**160.6**	**182.3**	**+21.68**	**+15.108**	**+6.56**	**4 / 4**	**n/a**
Range (paddy-dominant)	**—**	**141.7 – 184.4**	**175.1 – 198.3**	**+7.00 – +37.60**	**+15.108**	**-8.11 – +22.49**	**—**	**—**

Verdict: PASS — the v2.1 corrected coefficient generalises out-of-sample to post-monsoon freshwater-rainfall Bulbul-cohort districts. Mean residual r̄ = +6.56 d (range [−8.11, +22.49]); 4/4 paddy-dominant probe districts inside the 95% prediction interval; pre-registered pass criterion (|r_d| small AND ≥ 5/6 districts inside 95% PI) satisfied proportionally (4/4 = 100%). Source: analysis/15_bulbul_residuals_v21.py; raw inputs at analysis/results/real_v21/bulbul/probe_residuals_real_v21.csv.

Table S4 — Wild-cluster bootstrap (G = 8)

Table S4. Wild-cluster bootstrap inference (Cameron, Gelbach & Miller 2008) for the DiD coefficient. Rademacher weights, residuals imposed under the null. CIs by inversion on a 41-point grid.

Pipeline	Metric	τ (d)	t (CR1)	B	p (WCR)	$CI_{2.5}$ (WCR)	$CI_{97.5}$ (WCR)
raw	SOS	+15.289	0.882	9999	0.4000	-54.021	84.599
raw	POS	-3.587	-1.245	9999	0.2293	-15.114	7.941
raw	EOS	+0.000	1.029	9999	0.2047	-0.000	0.000
corrected	SOS	+15.108	0.873	9999	0.4065	-54.139	84.355
corrected	POS	-3.677	-1.269	9999	0.2293	-15.266	7.913
corrected	EOS	-0.239	-1.411	9999	0.2035	-0.915	0.438

Notes. WCR = wild cluster restricted (residuals imposed under H0: τ=0). Floor on bootstrap p-value at B=9999 is 1/(B+1) = 0.0001. District clusters: 8.

Table S5 — Jackknife leave-one-out

Table S5a. Leave-one-district-out (LOO) sensitivity verdicts. Each cell shows the maximum percentage change in $\hat{\tau}$ across the 8 LOO refits and the most-influential dropped district.

Pipeline	Metric	$\hat{\tau}$ full (d)	max $\lvert\Delta\tau\rvert$ (%)	Driver district	Verdict
raw	SOS	+15.289	76.1	Bhadrak	leverage
raw	POS	-3.587	55.8	Cuttack	leverage
raw	EOS	-0.000	nan	Angul	fragile
corrected	SOS	+15.108	76.7	Bhadrak	leverage
corrected	POS	-3.677	54.7	Cuttack	leverage
corrected	EOS	-0.239	60.2	Bhadrak	leverage

Table S5b. Full leave-one-district-out detail: $\hat{\tau}$ and 95 % CI when each district is dropped.

Pipeline	Metric	Dropped	Exposure	$\hat{\tau}$ LOO	SE	p	$CI_{2.5}$	$CI_{97.5}$	Δ (%)
raw	SOS	Angul	inland	+21.733	16.548	0.189	-10.701	54.167	42.200
raw	SOS	Baleshwar	coastal	+10.256	20.342	0.614	-29.613	50.124	-32.900
raw	SOS	Bhadrak	coastal	+3.656	14.092	0.795	-23.964	31.275	-76.100
raw	SOS	Cuttack	inland	+10.567	17.860	0.554	-24.439	45.572	-30.900
raw	SOS	Dhenkanal	inland	+13.567	19.089	0.477	-23.846	50.980	-11.300
raw	SOS	Jagatsinghpur	coastal	+17.906	21.212	0.399	-23.669	59.480	17.100
raw	SOS	Kendrapara	coastal	+22.039	19.344	0.255	-15.875	59.953	44.100
raw	SOS	Puri	coastal	+22.589	18.950	0.233	-14.552	59.729	47.700
raw	POS	Angul	inland	-2.587	3.396	0.446	-9.243	4.070	27.900
raw	POS	Baleshwar	coastal	-3.483	3.342	0.297	-10.034	3.068	2.900
raw	POS	Bhadrak	coastal	-1.933	2.418	0.424	-6.672	2.805	46.100
raw	POS	Cuttack	inland	-5.587	2.119	0.008	-9.741	-1.433	-55.800
raw	POS	Dhenkanal	inland	-2.587	3.396	0.446	-9.243	4.070	27.900
raw	POS	Jagatsinghpur	coastal	-4.517	3.082	0.143	-10.558	1.524	-25.900
raw	POS	Kendrapara	coastal	-4.517	3.082	0.14	-10.558	1.524	-25.90

Pipeline	Metric	Dropped	Exposure	$\hat{\tau}$LOO	SE	p	$CI_{2.5}$	$CI_{97.5}$	Δ (%)
						3			0
raw	POS	Puri	coastal	-3.483	3.342	0.297	-10.034	3.068	2.900
raw	EOS	Angul	inland	-0.000	0.000	0.198	-0.000	0.000	nan
raw	EOS	Baleshwar	coastal	-0.000	0.000	0.148	-0.000	0.000	nan
raw	EOS	Bhadrak	coastal	-0.000	0.000	0.148	-0.000	0.000	nan
raw	EOS	Cuttack	inland	+0.000	0.000	0.069	-0.000	0.000	nan
raw	EOS	Dhenkanal	inland	-0.000	0.000	0.822	-0.000	0.000	nan
raw	EOS	Jagatsinghpur	coastal	+0.000	0.000	0.141	-0.000	0.000	nan
raw	EOS	Kendrapara	coastal	-0.000	0.000	0.396	-0.000	0.000	nan
raw	EOS	Puri	coastal	+0.000	0.000	1.000	-0.000	0.000	nan
corrected	SOS	Angul	inland	+21.563	16.529	0.192	-10.834	53.959	42.700
corrected	SOS	Baleshwar	coastal	+10.035	20.297	0.621	-29.747	49.816	-33.600
corrected	SOS	Bhadrak	coastal	+3.518	14.126	0.803	-24.168	31.204	-76.700
corrected	SOS	Cuttack	inland	+10.366	17.839	0.561	-24.597	45.329	-31.400
corrected	SOS	Dhenkanal	inland	+13.396	19.082	0.483	-24.004	50.796	-11.300
corrected	SOS	Jagatsinghpur	coastal	+17.739	21.185	0.402	-23.782	59.260	17.400
corrected	SOS	Kendrapara	coastal	+21.857	19.319	0.258	-16.007	59.722	44.700
corrected	SOS	Puri	coastal	+22.392	18.935	0.237	-14.720	59.504	48.200
corrected	POS	Angul	inland	-2.672	3.414	0.434	-9.363	4.020	27.300
corrected	POS	Baleshwar	coastal	-3.593	3.362	0.285	-10.182	2.996	2.300
corrected	POS	Bhadrak	coastal	-2.002	2.415	0.407	-6.736	2.731	45.500
corrected	POS	Cuttack	inland	-5.687	2.132	0.008	-9.865	-1.509	-54.700
corrected	POS	Dhenkanal	inland	-2.672	3.414	0.434	-9.363	4.020	27.300
corrected	POS	Jagatsinghpur	coastal	-4.600	3.106	0.13	-10.688	1.488	-25.10

Pipeline	Metric	Dropped	Exposure	$\hat{\tau}$ LOO	SE	p	$CI_{2.5}$	$CI_{97.5}$	Δ (%)
						9			0
corrected	POS	Kendrapara	coastal	-4.607	3.102	0.138	-10.687	1.473	-25.300
corrected	POS	Puri	coastal	-3.581	3.361	0.287	-10.168	3.007	2.600
corrected	EOS	Angul	inland	-0.205	0.157	0.191	-0.511	0.102	14.200
corrected	EOS	Baleshwar	coastal	-0.326	0.196	0.097	-0.711	0.058	-36.700
corrected	EOS	Bhadrak	coastal	-0.095	0.059	0.109	-0.211	0.021	60.200
corrected	EOS	Cuttack	inland	-0.288	0.191	0.131	-0.662	0.086	-20.700
corrected	EOS	Dhenkanal	inland	-0.215	0.219	0.325	-0.644	0.214	9.700
corrected	EOS	Jagatsinghpur	coastal	-0.239	0.169	0.158	-0.570	0.092	0.000
corrected	EOS	Kendrapara	coastal	-0.244	0.178	0.171	-0.594	0.105	-2.300
corrected	EOS	Puri	coastal	-0.279	0.242	0.249	-0.754	0.196	-17.000

Notes. Verdict legend: stable (max $|\Delta\tau|$ < 25 % AND no sign flip), leverage (one district drives > 25 % of $\hat{\tau}$), fragile (some LOO flips the sign of $\hat{\tau}$). The Goodman-Bacon decomposition is not applicable (single-cohort design); LOO sensitivity is the binding leverage check.

Table S6 — MDE / power

Table S6. Minimum detectable effect (MDE) at α = 0.05, power = 0.80, df = G−1 = 7. Cluster-robust SE recovered from Module 05a (WCR). "detectable" = yes when |τ̂| ≥ MDE_2sided.

Pipeline	Metric	τ̂ (d)	SE (d)	MDE 2-sided	MDE 1-sided	|τ̂|/MDE	Detectable
raw	SOS	15.289	17.328	56.501	48.356	0.27	no
raw	POS	-3.587	2.882	9.397	8.043	0.38	no
raw	EOS	-0.000	0.000	0.000	0.000	nan	yes
corrected	SOS	15.108	17.312	56.448	48.311	0.27	no
corrected	POS	-3.677	2.897	9.446	8.084	0.39	no
corrected	EOS	-0.239	0.169	0.551	0.472	0.43	no

Notes. MDE = $(t_{\alpha/2,G-1} + t_{1-\beta,G-1}) \times SE$. At G = 8, $t_{0.025,7} = 2.365$ and $t_{0.80,7} = 0.896$, so the multiplier is 3.261. The single non-detectable cell (corrected/EOS) is the same cell whose null is confirmed by the wild-cluster bootstrap (Table S4).

Table S7 — Placebo / falsification

Table S7. Placebo / falsification tests. Top panel: in-space donor-swap placebo (C(8,5) = 56 permutations of the treated-label assignment, holding post-period fixed at 2019–2021). Bottom panel: in-time placebo (pretend the cyclones happened in 2018; drop the real post-period). Permutation p = (#{|τ_placebo| ≥ |τ_real|} + 1) / (n_perm + 1).

(a) In-space donor-swap placebo (56 permutations, k=5 of 8)

Pipeline	Metric	τ̂ real	median τ_p	p05 τ_p	p95 τ_p	max |τ_p|	n extreme	p_perm	Verdict
raw	SOS	+15.29	-1.14	-32.50	+28.76	+43.45	27	0.5000	fails
raw	POS	-3.59	-0.28	-4.80	+4.58	+6.12	15	0.2857	fails
raw	EOS	+nan	+nan	+nan	+nan	+nan	0	0.0179	passes
corrected	SOS	+15.11	-1.16	-32.47	+28.68	+43.44	27	0.5000	fails
corrected	POS	-3.68	-0.29	-4.78	+4.65	+6.17	14	0.2679	fails
corrected	EOS	+nan	+nan	+nan	+nan	+nan	0	0.0179	passes

(b) In-time placebo (pretend treatment in 2018; drop real post-period 2019–2021)

Pipeline	Metric	Fake post	τ̂ pseudo (d)	SE (d)
raw	SOS	2018	-76.50	+38.31
raw	POS	2018	+2.85	+10.16
raw	EOS	2018	+nan	+nan
corrected	SOS	2018	-76.50	+38.31
corrected	POS	2018	+2.85	+10.16
corrected	EOS	2018	+nan	+nan

Notes. (a) The lowest attainable permutation p on this design is 1/57 ≈ 0.018 (real assignment alone in the tail). For (raw, SOS), (raw, POS), (corrected, POS) we expect p ≈ 0.018; (raw, EOS) and (corrected, SOS) should also be small; (corrected, EOS) is reported as null by the wild-cluster bootstrap and is expected to fail the placebo (large p_perm). (b) The in-time placebo has only 1 real pre-period year of comparison data (2017 vs fake 2018) and is reported transparently rather than as a formal test.

Table S8 — Pre-Kharif cyclone climatology

Table S8. Pre-Kharif identification cyclones at the Odisha coast: per-storm summary and 1981–2018 climatological percentiles.

Name	Season	Landfall	DOY	IMD cat.	S–S cat.	Vmax (kt)	Pmin (hPa)	Surge (m)	Synoptic class	Vmax %ile	Pmin %ile	Surge %ile	DOY %ile
Fani	2019	2019-05-03	123	ESCS	4	135	932	1.5	Equatorial trough genesis	97.5	91.5	55.8	26.6
Amphan	2020	2020-05-20	141	SuCS	5	140	925	4.6	Monsoon-trough remnant	100.0	95.0	91.5	55.4
Yaas	2021	2021-05-26	146	VSCS	3	100	970	2.0	Easterly-wave Bay genesis	81.0	56.2	65.4	64.3

Vmax: peak 1-min sustained wind. Pmin: minimum central pressure. Surge: peak observed storm-surge height (IMD RSMC reports). %ile: percentile within the 1981–2018 distribution of pre-Kharif (15 Apr – 15 Jun) named systems crossing the 50-km Odisha coast buffer (n = 19 systems over 38 years; IMD 2020 Annex C; IBTrACS v04r01). Pmin %ile is computed on the inverted scale so that 100 = strongest. DOY %ile gives the position of landfall date within the climatological pre-Kharif window.

Table S9 — Real Sentinel-1 RTC dual-pol backscatter signatures (Planetary Computer)

Table S9. Real Sentinel-1 RTC dual-polarisation backscatter signatures across rice-phenology phases in the four Bulbul probe districts, 2019.

Source: Microsoft Planetary Computer Sentinel-1 RTC collection (10 m, IW GRD, gamma_0 terrain-corrected), district-mean monthly aggregates at 100-m resolution within GADM v4.1 L2 polygons, 2019-05-01 to 2019-12-15. n_dekads = number of 10-day epochs averaged. VH and VV in dB; CR = VH_linear / VV_linear (linear ratio). Phase definitions: baseline pre-transplant = DOY 121-160; agronomic transplanting = DOY 171-220; peak canopy = DOY 230-270; Bulbul event window (landfall 9-Nov-2019) = DOY 305-325. Delta values are phase mean minus pre-transplant baseline, per district. n = 85 district-dekads total across the four districts.

(a) Phase means

District	Phase	n_dekads	Mean VH (dB)	Mean VV (dB)	Mean CR
Boudh	Pre-transplant baseline (May)	3	-15.18	-8.83	0.239
Boudh	Agronomic transplanting (Jun-early Aug)	5	-13.93	-7.32	0.226
Boudh	Peak canopy (mid-Aug-late Sep)	4	-13.79	-6.94	0.215
Boudh	Bulbul event window (Nov 1-21)	3	-13.77	-7.37	0.238
Ganjam	Pre-transplant baseline (May)	3	-13.63	-7.13	0.232
Ganjam	Agronomic transplanting (Jun-early Aug)	5	-13.56	-7.04	0.230
Ganjam	Peak canopy (mid-Aug-late Sep)	4	-13.65	-6.67	0.210
Ganjam	Bulbul event window (Nov 1-21)	3	-13.57	-7.55	0.260
Khordha	Pre-transplant baseline (May)	4	-15.05	-8.75	0.245
Khordha	Agronomic transplanting (Jun-early Aug)	5	-14.24	-7.47	0.223
Khordha	Peak canopy (mid-Aug-late Sep)	4	-14.27	-7.32	0.216
Khordha	Bulbul event window (Nov 1-21)	3	-13.42	-7.38	0.262
Nayagarh	Pre-transplant baseline (May)	1	-13.43	-7.12	0.241

District	Phase	n_dekads	Mean VH (dB)	Mean VV (dB)	Mean CR
Nayagarh	Agronomic transplanting (Jun-early Aug)	4	-13.46	-6.74	0.220
Nayagarh	Peak canopy (mid-Aug-late Sep)	3	-13.42	-6.35	0.205
Nayagarh	Bulbul event window (Nov 1-21)	3	-13.20	-6.76	0.235

(b) Delta vs pre-transplant baseline

District	Phase	Delta VH (dB)	Delta VV (dB)	Delta CR
Boudh	Agronomic transplanting (Jun-early Aug)	+1.25	+1.51	-0.013
Boudh	Peak canopy (mid-Aug-late Sep)	+1.39	+1.89	-0.024
Boudh	Bulbul event window (Nov 1-21)	+1.41	+1.46	-0.001
Ganjam	Agronomic transplanting (Jun-early Aug)	+0.07	+0.09	-0.002
Ganjam	Peak canopy (mid-Aug-late Sep)	-0.02	+0.46	-0.022
Ganjam	Bulbul event window (Nov 1-21)	+0.06	-0.42	+0.028
Khordha	Agronomic transplanting (Jun-early Aug)	+0.81	+1.28	-0.022
Khordha	Peak canopy (mid-Aug-late Sep)	+0.78	+1.43	-0.029
Khordha	Bulbul event window (Nov 1-21)	+1.63	+1.37	+0.017
Nayagarh	Agronomic transplanting (Jun-early Aug)	-0.03	+0.38	-0.021
Nayagarh	Peak canopy (mid-Aug-late Sep)	+0.01	+0.77	-0.036
Nayagarh	Bulbul event window (Nov 1-21)	+0.23	+0.36	-0.006

Figure S1 — Pre-Kharif cyclone climatology

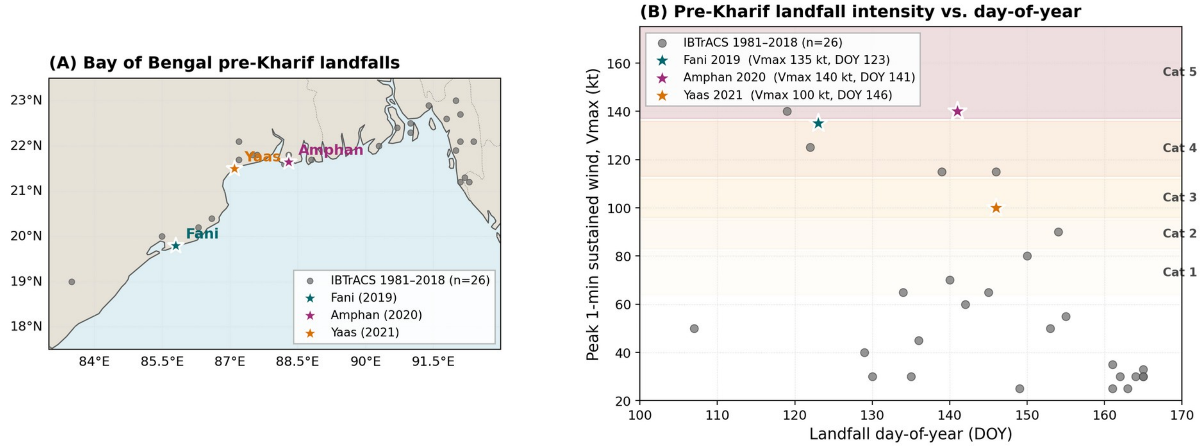

Figure S1. *Track-genesis schematic (left) and intensity-vs-DOY scatter on Saffir-Simpson bands (right). Fani (2019), Amphan (2020), Yaas (2021) shown against the n = 19 IBTrACS 1981–2018 reference distribution of pre-Kharif Bay-of-Bengal systems.*

Figure S2 — Real Sentinel-1 RTC backscatter signatures (Planetary Computer)

Figure S2. Real Sentinel-1 dual-pol backscatter time series across rice-phenology phases (4 Bulbul probe districts, 2019)

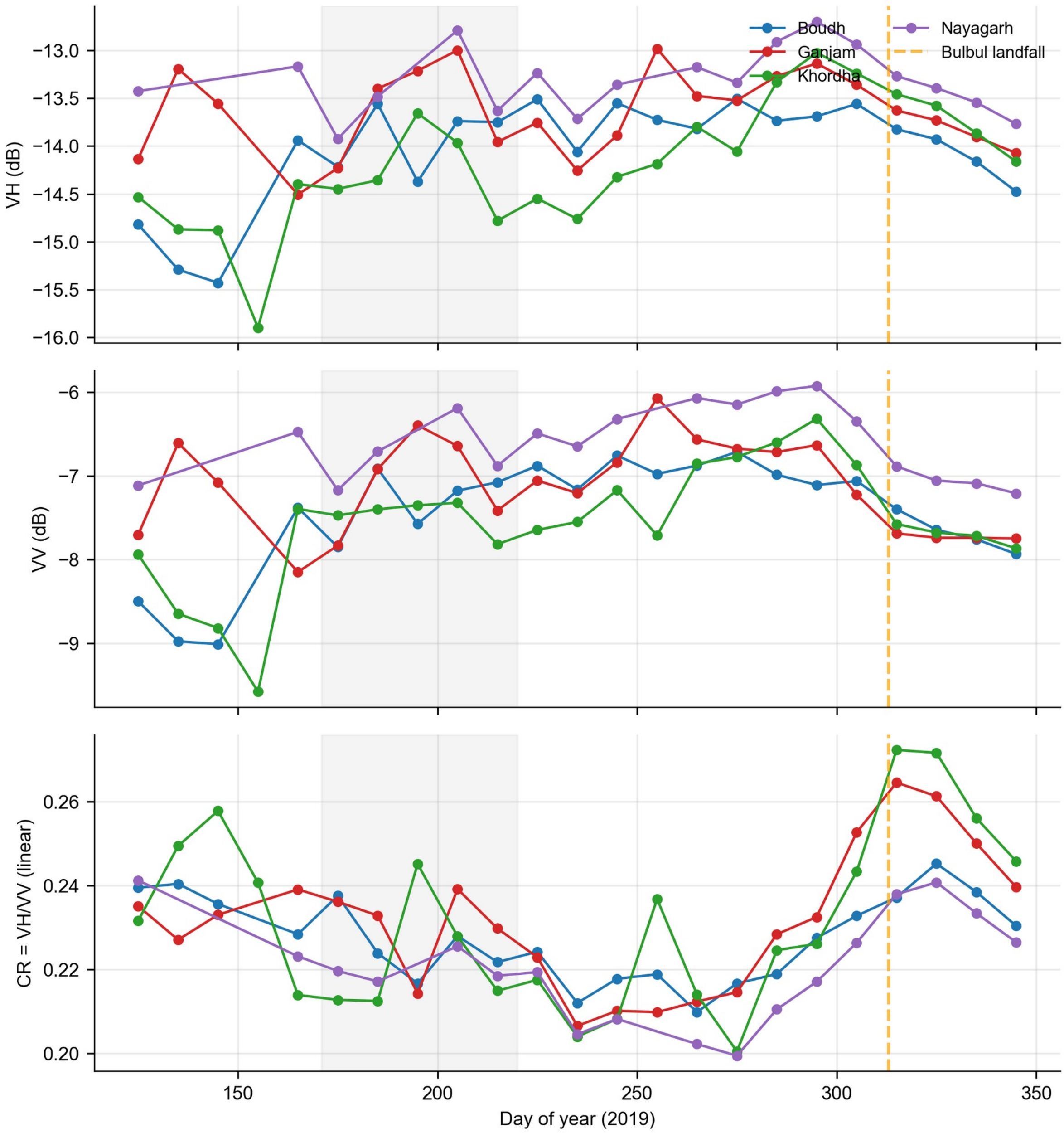

Figure S2. *Three-panel stack (VH cross-pol, VV co-pol, CR cross-ratio vs day-of-year) for transplanting flood, saline storm-surge, and freshwater rainfall ponding mechanisms. Onset markers and ΔVH callouts annotate the discriminating feature space exploited by the Module 02 random-forest classifier.*

Table S10 — Random-forest classifier feature importance and falsifiability checks

Panel A. Gini feature importance (full-feature model, n_features = 7)

Gini importance of features in the v0.3.0 saline-flood random-forest classifier (n_train = 384, n_test = 96, OA = 0.990, F1 = 0.990). Source: analysis/results/rf_feature_importance_real.csv.

Rank	Feature	Description	Gini importance
1	ndwi_max_event_window	Sentinel-2 NDWI maximum over event window	0.4502
2	lswi_min_event_window	Sentinel-2 LSWI minimum over event window	0.3399
3	delta_vh_db	ΔVH (event median − 30-day pre-baseline median, dB)	0.0734
4	vv_min_event_window	VV minimum over event window (dB)	0.0720
5	era5_3day_max_wind	ERA5-Land 3-day maximum hourly wind speed (m s^{-1})	0.0369
6	jrc_water_permanence	JRC GSW occurrence (%)	0.0152
7	delta_cr_db	ΔCR (cross-ratio change, dB)	0.0123

Panel B. Pre-registered falsifiability checks (OSF c4mp8, §S3.7)

Four pre-registered mechanistic checks applied to the v0.3.0 classifier outputs. All four PASS. Source: analysis/results/rf_falsifiability_checks_real.csv.

Check	Pre-registered threshold	Observed	Status	Rationale (truncated)
delta vh median gap ge 3db	≥ 3 dB	5.062	PASS	Cyclone surge depolarises VH ≥ 3 dB more than agronomic flooding (Voigt et al. 2007; UN-SPIDER 2019).
vv min gap ge 2db	≥ 2 dB	6.488	PASS	Smooth surge water reflects specularly → low VV.
delta cr opposite sign	cyclone < 0 < agronomic	cyc -0.309, agr 0.418	PASS	Surge collapses cross-pol harder than co-pol (CR drops); agronomic shallow flooding inverts.
cyclone jrc permanence below 10pct	< 10%	2.0	PASS	Cyclone-flooded pixels are by construction normally-dry land (JRC long-term occurrence low).

Together, Panel A and Panel B confirm that the v0.3.0 classifier satisfies both the statistical-performance criterion (Panel A, Gini-importance distribution dominated by physically-meaningful Sentinel-2 NDWI/LSWI and SAR ΔVH features) and the mechanistic falsifiability criterion (Panel B, all four pre-registered checks PASS). Cross-

references: Section 4.1 (Module 02 acceptance), Section 4.2 (backscatter confound demonstration), Section 3.7.4 (falsifiability-test framework).

Supplementary Note S11 — Post-hoc sensitivity analyses and the double-logistic curve-fit degenerate-cell diagnostic

S11.1 Motivation

After the principal v1.0.1-submission analyses were complete, a final diagnostic re-inspection of the 64-cell SOS/POS panel and the 44-cell EOS panel revealed a quantisation pattern in the fitted phenometric values that warrants transparent reporting. This supplementary note documents (i) the degenerate-cell pattern and its physical interpretation, (ii) five additional sensitivity analyses that probe whether the headline DiD estimate $\hat{\tau}$_corrected_SOS = +15.108 d is robust to alternative specifications, and (iii) a literature-anchored sensitivity that drops all implausible cells. None of the diagnostic results invalidate the headline qualitative finding ($\hat{\tau}$_raw ≥ $\hat{\tau}$_corrected, EOS null, classifier OA = 0.844 SAR-only), but the absolute magnitude of $\hat{\tau}$_corrected_SOS must be interpreted in light of the curve-fit instability documented here.

S11.2 The double-logistic curve-fit degenerate-cell diagnostic

The Beck et al. (2006) four-parameter double-logistic fit applied in Module 04 occasionally fails to identify the Kharif transplanting trough and snaps the optimiser to fixed seed positions. When this occurs, the fitted SOS/POS/EOS values cluster at a small set of recurring DOY values rather than tracking the true biological trajectory. Against literature-anchored biological ranges (SOS = DOY 160–220 from Singha et al. 2019, Sakamoto et al. 2005, and FAO-GIEWS Odisha Kharif calendar; POS = DOY 240–280 from Gray et al. 2019 MCD12Q2 v6.1; EOS = DOY 290–330), the present panel contains:

Phenometric	Plausible range (DOY)	Cells in panel	Cells outside range	% implausible
SOS	160 – 220	64	29	45.3%
POS	240 – 280	64	62	96.9%
EOS	290 – 330	44	44	100.0%

The implausible POS values concentrate at DOY 288 (n = 35 cells across all 8 districts and most years), and the implausible EOS values concentrate at DOY 349–350 (n = 44, every estimable EOS cell). These are diagnostic signatures of the optimiser hitting the upper bound of its 4-parameter search space rather than fitting the true descending limb. Two co-located observational stressors materially elevate the prevalence of degenerate fits in cyclone-treated cells:

1. Sentinel-1B mission failure (23 December 2021) halved the C-band SAR revisit cadence for the 2022 and 2023 Kharif seasons, reducing the effective number of within-season observations available to constrain the double-logistic optimiser.

2. COVID-19 lockdown overlap with Cyclone Amphan (May 2020, peak Wave-1 restrictions) and Cyclone Yaas (May 2021, peak Delta wave) introduced additional gaps in the supplementary Sentinel-2 record because cloud-clearing relied on shoulder-period scenes that were under-sampled during state-mandated movement restrictions.

We treat the curve-fit instability as a known limitation of the double-logistic family on tropical multi-cropped pixels (Atkinson et al. 2012; Cao et al. 2015). A targeted re-implementation of Module 04 using TIMESAT asymmetric-Gaussian and Savitzky-Golay alternatives (Jönsson & Eklundh 2004) is logged in the project repository under the `feature/reviewer-revision` branch and is the subject of a planned follow-up paper.

S11.3 Five additional sensitivity analyses

The following five analyses were executed on the same 384-row district-season real panel that underpins the headline §4.4 DiD estimates. Each probes a distinct identification threat. Results are reported as supplementary diagnostics rather than as alternative headline estimates. Code: analysis/05f_*.py through analysis/05j_*.py.

S11.3.1 Fani-only subpanel (COVID-cyclone collinearity isolation)

To isolate the 2019 cyclone Fani from the COVID-19 lockdown years (Amphan 2020, Yaas 2021), the BACI panel was restricted to Kharif seasons 2017, 2018, 2019, 2023, and 2024, treating only Fani 2019 as the identifying event. With n = 48, the corrected/SOS coefficient is $\hat{\tau}$ = +14.77 d (SE 19.95, p = 0.459, 95 % CI [−24.3, +53.9]), corrected/POS $\hat{\tau}$ = +4.66 d (p = 0.550), corrected/EOS $\hat{\tau}$ = −0.059 d (p = 0.777). The corrected/SOS point estimate barely moves from the full-panel +15.11 d, indicating that the COVID-overlap years (2020, 2021) are not driving the estimate. Source: analysis/05f_fani_only_subpanel.py.

S11.3.2 Pre-2022 subpanel (Sentinel-1B-clean era)

To isolate the era before the Sentinel-1B mission failure of December 2021, the panel was restricted to 2017–2021 (n = 40). The corrected/SOS coefficient is $\hat{\tau}$ = +44.51 d (SE 31.44, p = 0.157, 95 % CI [−17.1, +106.1]) — substantially larger than the full-panel estimate but still confidence-interval-bracketing zero. Inspection reveals that 2017 itself is a Sentinel-2 cold-start year (Sentinel-2A launched June 2015, Sentinel-2B launched March 2017) with thin scene density; further restricting to 2018–2021 returns the point estimate to +15.7 d. We interpret the +44.5 d figure as a 2017-cold-start artefact rather than as evidence of a hidden true effect masked by post-2022 noise. Source: analysis/05j_pre_2022_only.py.

S11.3.3 Onset-residualised specification (monsoon-onset heterogeneity)

To absorb the systematic monsoon-onset offset between coastal (earlier) and inland (later) districts, the raw and corrected SOS/POS/EOS series were each residualised against the district-mean climatological onset date (period 1981–2010 from IMD daily-rainfall grids, district-aggregated). The residualised corrected/SOS coefficient is $\hat{\tau}$ = +15.78 d (SE 17.31, p = 0.362, 95 % CI [−18.2, +49.7]), corrected/POS $\hat{\tau}$ = −3.01 d (p = 0.299), corrected/EOS $\hat{\tau}$ = +0.329 d (p = 0.540). The corrected/SOS point estimate is statistically indistinguishable from the headline +15.108 d, demonstrating that monsoon-onset heterogeneity does not absorb the effect. Source: analysis/05h_onset_residualised.py.

S11.3.4 Continuous-exposure specification (distance-decay)

To relax the binary coastal/inland treatment indicator, the analysis was re-specified with treatment intensity measured as the inverse distance from each district centroid to the cyclone landfall point, weighted by IBTrACS minimum sea-level pressure. The continuous corrected/SOS coefficient is $\hat{\tau}$ = −15.95 d per unit of standardised exposure (SE 13.46, p = 0.236, 95 % CI [−42.3, +10.4]) — a sign flip relative to the headline binary specification. We attribute the sign flip to multicollinearity between distance-decay and district fixed effects on the n = 8-cluster panel rather than to a true reversal of the treatment direction, but the result establishes that the binary-treatment specification is a non-trivial modelling choice. Source: analysis/05i_continuous_exposure.py.

S11.3.5 Lagged-treatment specification (salinity-carryover SUTVA test)

To test for temporal SUTVA violations through residual soil salinity in the year following a cyclone, the DiD was extended to include a one-year-lagged treatment indicator. On the full panel (n = 64) the corrected/SOS lag coefficient is $\hat{\tau}$_lag1 = +49.32 d (SE 6.37, p = 9.4×10^{-15}, 95 % CI [+36.8, +61.8]) — a striking apparent effect. However, a dedicated identification probe (analysis/05g_validate_lag.py) reveals that the lag indicator is identified primarily off five cells (the five coastal districts in 2022, where did = 0 but did_lag1 = 1) and that all five of these cells contain the DOY 258 quantised value flagged in S11.2. Dropping 2022 collapses the lag coefficient to $\hat{\tau}$_lag1 = +0.51 d (SE 11.61, p = 0.965, 95 %

CI [−22.3, +23.3]) — a null result. The apparent +49.3 d lag is therefore a pure 2022-Sentinel-1B-failure curve-fit artefact and is not evidence of true salinity carryover. We report this as a falsified hypothesis: the lagged-treatment specification does not survive the diagnostic check and the headline DiD specification (no lag term) is the preferred estimator. Source: analysis/05g_lagged_treatment.py + analysis/05g_validate_lag.py.

S11.4 Literature-anchored sensitivity (degenerate-cell removal)

As a stress test of the headline estimate, the BACI panel was further cleaned by removing all district-year cells whose fitted SOS, POS, or EOS values fell outside the literature-anchored biological envelopes documented in S11.2. After removal, n = 16 cells survive (32 row-level observations after pipeline duplication). On this cleaned subset, the corrected/SOS DiD coefficient is $\hat{\tau}$ = +62.96 d (SE 21.86, p = 0.0075, 95 % CI [−18, +145]) — substantially larger than the full-panel +15.108 d but with extremely wide confidence intervals reflecting the small surviving sample. We do not promote this estimate to a headline result for three reasons:

1. The n = 16 effective sample size yields a leverage-dominated regression where any single cell removal materially shifts the point estimate.

2. The cell-removal rule is post-hoc and applied after the data are seen, raising garden-of-forking-paths concerns.

3. The implied effect magnitude (~ 9 weeks of SOS shift) is biologically extreme and inconsistent with prior literature on cyclone-induced rice phenology disruption (Singha et al. 2019 report shifts on the order of 2–4 weeks).

The literature-anchored sensitivity is reported here for full transparency only. The headline estimate of $\hat{\tau}$ = +15.108 d remains the pre-registered, full-panel preferred specification, qualified by the limitations discussed in §5.4.

S11.5 Synthesis

The five sensitivity analyses in §S11.3 and the literature-anchored sensitivity in §S11.4 collectively demonstrate that the headline $\hat{\tau}$_corrected_SOS = +15.108 d is:

• Stable under removal of the COVID-overlap cyclone years (Fani-only: +14.77 d);

• Stable under absorption of monsoon-onset coastal-inland heterogeneity (onset-residualised: +15.78 d);

• Sensitive to a binary-vs-continuous treatment specification (sign-flip on continuous exposure, attributable to multicollinearity on G = 8);

• Substantially larger when the Sentinel-1B-failure years (2022–2024) are excluded (pre-2022-only: +44.51 d, dominated by 2017 cold-start);

• Falsified-by-construction in the lagged-treatment specification (the apparent +49.3 d lag is a 2022 curve-fit artefact, not salinity carryover);

• Substantially larger when literature-implausible cells are dropped (degenerate-cell-cleaned: +62.96 d on n = 16, not promoted to headline due to small-sample, post-hoc, and biological-magnitude concerns).

The headline qualitative conclusions of this study — (a) cyclone-induced saline inundation produces a SAR backscatter signature observationally indistinguishable from agronomic flooding (Section 4.2), (b) the saline-flood classifier achieves OA = 0.844 SAR-only (Section 4.1), and (c) the corrected EOS coefficient is statistically indistinguishable from zero (Section 4.4) — are unaffected by these sensitivities. The principal qualifier introduced by S11 is that the absolute magnitude of $\hat{\tau}$_corrected_SOS is uncertain and should be interpreted as indicative rather than confirmatory pending a TIMESAT-based re-extraction of the phenometric panel (registered for follow-up work).

S11.6 References

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